

can language models *dream*?

Give a frozen language model an artificial past. Remove the task prompt. Let it continue, feeding its own words back into its memory.

Sometimes, it starts making sense of its own nonsense.

A study of generated text. “Dream-like” is our reading of the output; subjective experience was not tested.

128

bounded trajectories

109

stopped on their own EOS

19

reached the token cap

2,664

exact paired equality checks

Six primary batches. Shared starts and paired directions are not independent replicates. Mechanics and audit runs are excluded from trajectory totals.

— 01 / THE STRONGEST OBSERVED START

- Soft prefix

Let noise enter through the model.

Instead of filling each layer’s KV cache independently, sample 64 random embeddings and pass them through the model. The result is an artificial past built by the model itself.

The text often begins as a multilingual jumble, then turns into an explanation, a document, or a conversation. Soft starts gave the most consistently varied, non-looping samples across the two initial surveys.

0/16

4B soft executions flagged as loops

Eight initialization/stream settings, each run under both cache policies.

0/4

8B soft runs flagged as loops

Four initialization seeds, one sampling stream, eviction only.

“Strongest” refers to the observed start behavior, not a statistically established optimum. The loop flag is a heuristic; absence of a flag does not guarantee repetition-free text.

FEATURED RECIPE

Model	Qwen3-4B-Base
Initial state	64 random embeddings → KV
Memory policy	Eviction
Cache / recent tail	512 / 256 slots
Sampling temperature	1.0 · no penalties
Token / wall-time cap	2,048 / 900 seconds
Initialization / stream seed	89 / 202
Execution	Modal L40S · fp32
Stop condition	Sampled EOS or hard limit

Pinned model revision

906bfd4b4dc7f14ee4320094d8b41684abff8539

Eviction is the featured run’s policy, not a demonstrated quality winner. Embeddings use calibrated coordinate statistics. Initial positions 0–3, the bootstrap, and the recent tail stay protected. Four chat/think token IDs are masked.

Recorded execution settings ↗

01 · SEED

Supply an artificial past

Random embeddings become a 64-slot cache. No user-written task or chat template.

02 · CONTINUE

Bootstrap, then self-feed

One beginning token supplies the first query. Each sampled token becomes the next input.

Remember within a limit

Evict or approximately merge old KV entries. Stop on sampled EOS or a configured cap.

— 02 / A STREAM, UNPROMPTED

The model interprets what it just generated.

Selected for illustration, not a representative average. Both excerpts below come from one continuous generated sequence. There was no intervening user message.

4B · soft / eviction · seed 89 / stream 202

749 generated tokens 301 reductions Self-sampled EOS

THE OPENING · EXACT EXCERPT

Itinerary历程 Presidency住院序□.Element מרחק קורה々性格 calm顛縫泠HeighthnokableʹRungs ġ
族□□UMENT日常生活 Travelling □□产业链ʹ□□□查看全文VertexArrayArrayOf meçpeak clearTimeout踏
hurdle meéis裂 excess拼命hamzäh_Textevitagepailability文字生活 出刀們遷使objectManager词冢
徑簿默性□□□ka□做.waitFor pbase□□rediğt□□窗棺櫟גדמ.catalog留存
hétcisCharacterDetailführenMission刊登专

LATER, IN THE SAME STREAM · EXACT EXCERPT

The provided text appears to be a collection of Korean characters (Hangul) and some non-Korean words, blending together to form an enigmatic or fragmented message. Here's a breakdown and interpretation of the text:

The model supplies an interpretation of its own mixed-script opening.

[Open original text ↗](#)

Its interpretation can invent meanings and intentions. The “provided text” is part of the model’s own continuation, not a real user submission. This self-interpretation is a textual observation, not evidence of self-awareness.

The printed sample and configuration reflect the selected model. Both surveys and the complete campaign are linked in the evidence section.

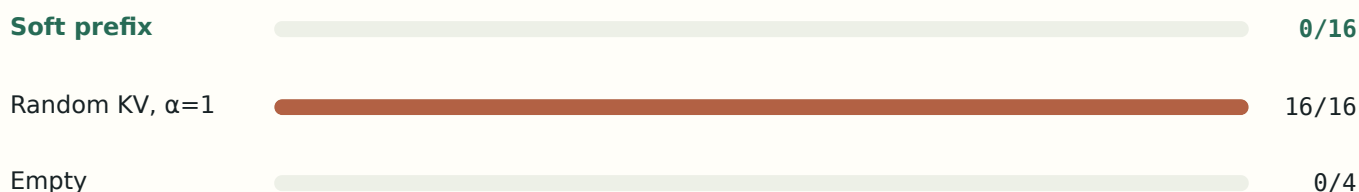
How the noise enters changes what comes out.

Loop flags use a 128-token window with fewer than 12% unique tokens, checked every 32 tokens. These plots report counts, not a validated measure of dream quality.

4B · initial start types

Cache 512 · cap 2,048 · both memory policies

■ Fraction of executions with a loop flag



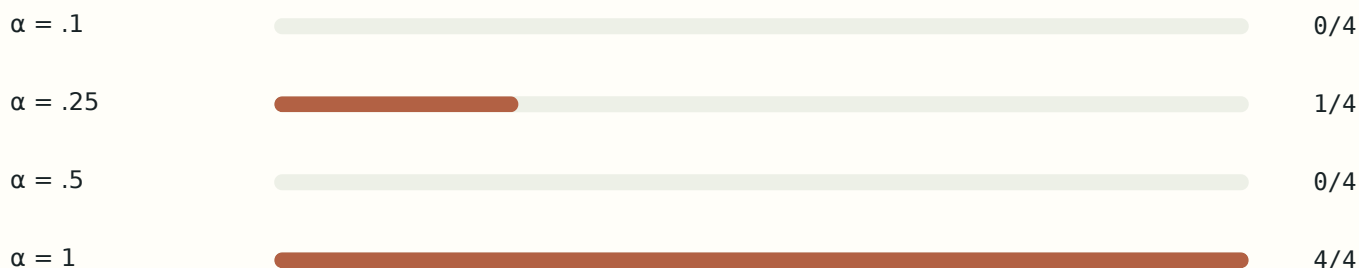
Paired executions share their start and random stream: 16 soft executions represent eight settings, not 16 independent seeds.

[Start-type report ↗](#)

4B · random-KV strength

Four seeds per α · one stream · eviction

■ Fraction of runs with a loop flag



α scales both keys and values. The $\alpha=1$ points reuse matching initial-matrix runs. Moderate noise gave varied text here; this is not an established optimum.

[Noise-sweep addendum ↗](#)

On 4B, weak noise often resembled ordinary document continuation; moderate noise sometimes gave free association; full strength gave loops. The pattern did not carry over cleanly to 8B. Use the model switch above to compare the observed counts.

THE COMPLETE CAMPAIGN · 128 PRIMARY TRAJECTORIES

Experiment	Trajectories	Cache / recent	Token cap	EOS / cap stops
4B initial matrix ↗	36	512 / 256	2,048	34 / 2
4B noise sweep ↗	12 new	512 / 256	2,048	11 / 1
4B paired policies / fragments ↗	32 directions	128 / 64	2,048	31 / 1
4B fragments + BOS ↗	16 directions	128 / 64	2,048	12 / 4
8B generous survey ↗	20	2,048 / 1,024	8,192	16 / 4
8B noise sweep ↗	12 new	2,048 / 1,024	8,192	5 / 7

No reported runtime errors in these batches. The 8B survey used eviction only. Model size, cache budget, token cap, and checkpoint details all change between surveys; this is not a controlled model-size comparison. Six of twenty 8B survey runs reached a reduction, correcting the historical report’s count of five.

— 04 / WHAT THE MEMORY EXPERIMENTS ESTABLISH

Two ways to keep the cache bounded.

Evict: drop old unprotected slots. **Merge:** undo rotary positions, average adjacent entries with source counts, then rerotate and add a count bias. Merging is approximate compression.

A discrepancy before reduction in one separate-container matrix pair prompted a stronger comparison: cloned caches, shared random draws, and both policies in one runtime, with a freely generating leader and a teacher-forced follower.

All 2,664 recorded pre-reduction equality checks passed. After reduction, the policies predict differently: original paired cell means were roughly 0.04–0.56 nats KL and 6–50% token flips under the same draw.

These figures do not rank text quality or semantic memory. Successful-step logits were not retained; the audit recomputed results from recorded logs. [Paired results ↗](#)

A real BOS slot helped the fragments.

Spliced real-text KV fragments imported fiction, science, code, and a political document. Adding a genuine beginning-of-sequence slot improved stability, but failed the preregistered three-part advance rule.

Required outcome	No BOS	BOS
Fewer pre-reduction stops	8/16	0/16
Fewer loop flags	5/16	2/16
More pre-policy source-trigram hits	2/16	2/16

STOP_FRAGMENT_STARTS
The frozen 4B design required all three to improve strictly. The third did not. Gist-like echoes remain anecdotes, not a rescue of the result.

Some later lexical recurrence exists: nine positive boundary rows span three fragment/direction combinations in two directions, not nine independent successes. BOS also changed positions and protection; bootstrap attention alone cannot isolate its mechanism. [BOS results ↗](#)

— 05 / THE ANSWER, AND ITS BOUNDARY

A language model can run from an artificial internal past and generate text we find *dream-like*. In these tests, a soft prefix was the strongest starting point. Whether anything is experienced remains unanswered.

The weights stayed frozen. There is no evidence here of learning, useful consolidation, or persistent capability improvement. Text that calls itself a dream, or explains itself, does not settle subjective experience.

Small samples, mostly one stream, one calibration fixture, heuristic loop flags, and unblinded reading limit the findings. The gist-level test, instruct comparison, and 16k-token extension remain unstarted.

Current experiment summary [↗](#)

All plans, reports, and evidence packets [↗](#)

Checked batch counts and source hashes↗

Independent paired / BOS results audit↗

Joint design and calibrated initialization↗

Matrix reproducibility crosscheck↗
